

Choosing and Evaluating AI

Sonja Aits

Lund University
2026-09-17



Define the problem

- What is the goal?
- What is the scope?
- How will you measure success?

Understanding project context

- How have others approached this?
- What resources and skills do you have?
- Operational requirements?

Making an initial plan

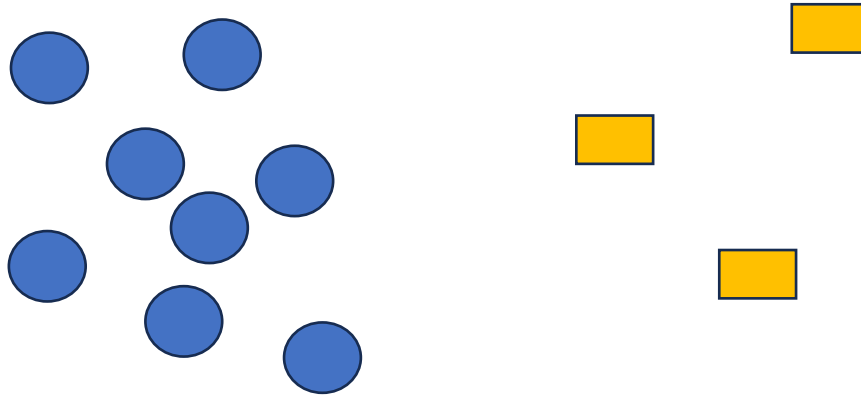
- What AI approach will you use?
 - AI task
 - Learning paradigm
 - Model types and architectures
 - Evaluation strategy

Break it down...

- Regression
- Clustering
- Classification
- Simplification
- Data generation
- Reasoning
- ...

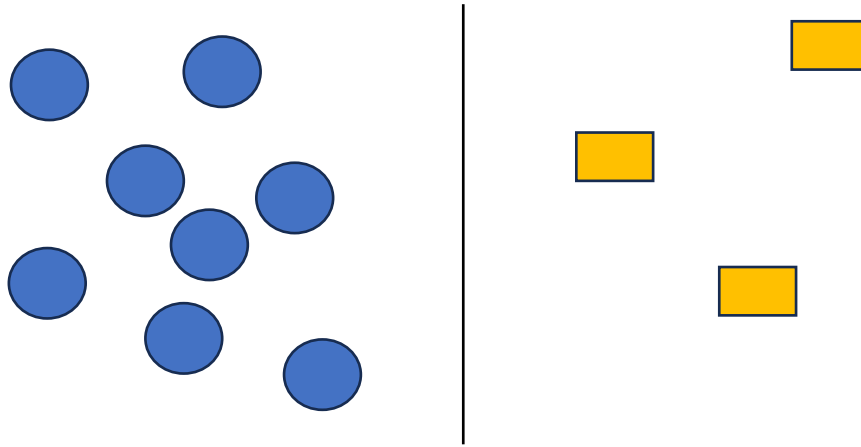
Many problems can be solved with different AI approaches

Finding faulty measurements
(Outlier detection)



Many problems can be solved with different AI approaches

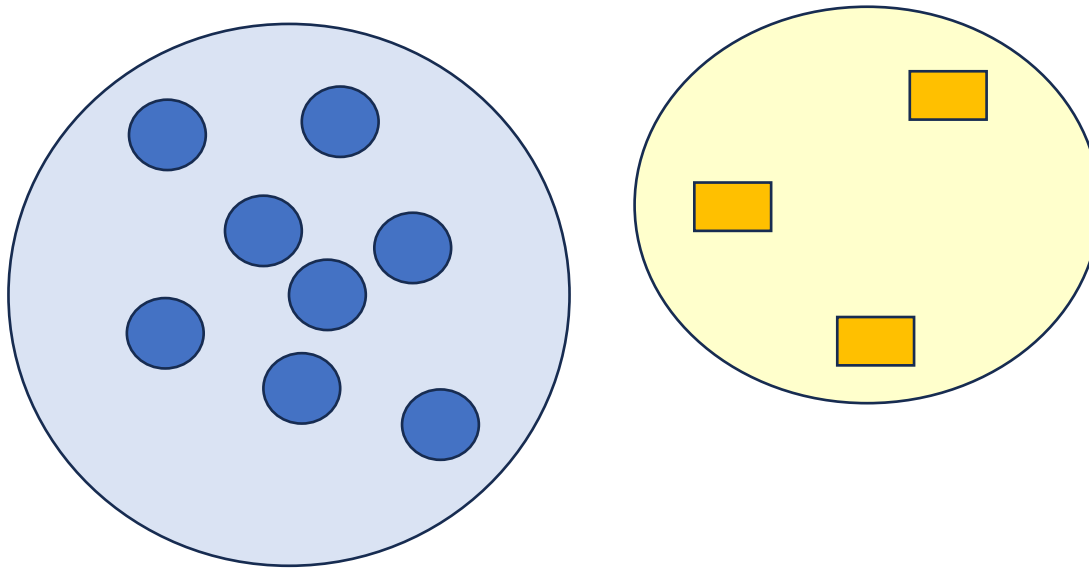
Finding faulty measurements
(Outlier detection)



Classification

Many problems can be solved with different AI approaches

Finding faulty measurements
(Outlier detection)



Clustering

Complex problems often require a combination of several AI core tasks

Targeted pesticide spraying



Classification: pest identification

Reasoning: pesticide selection, flight course plotting

Regression: amount of pesticide, rotor speed

AI comes in many shapes

Rule-based systems

Decision tree

K-nearest neighbor

K-means clustering

Linear regression

Convolutional neural network

Transformer

Recurrent neural network

Graph neural network

Autoencoder

Generative adversarial network

Classical machine learning – advantages compared to deep learning

- Higher explainability
- Often less data required
- Usually faster to train
- Require less computational resources
- Possibly superior performance with structured (tabular) data

BUT:

often insufficient for complex tasks with unstructured data

→ Deep learning

Exponential growth of parameters in notable AI systems

Our World
in Data

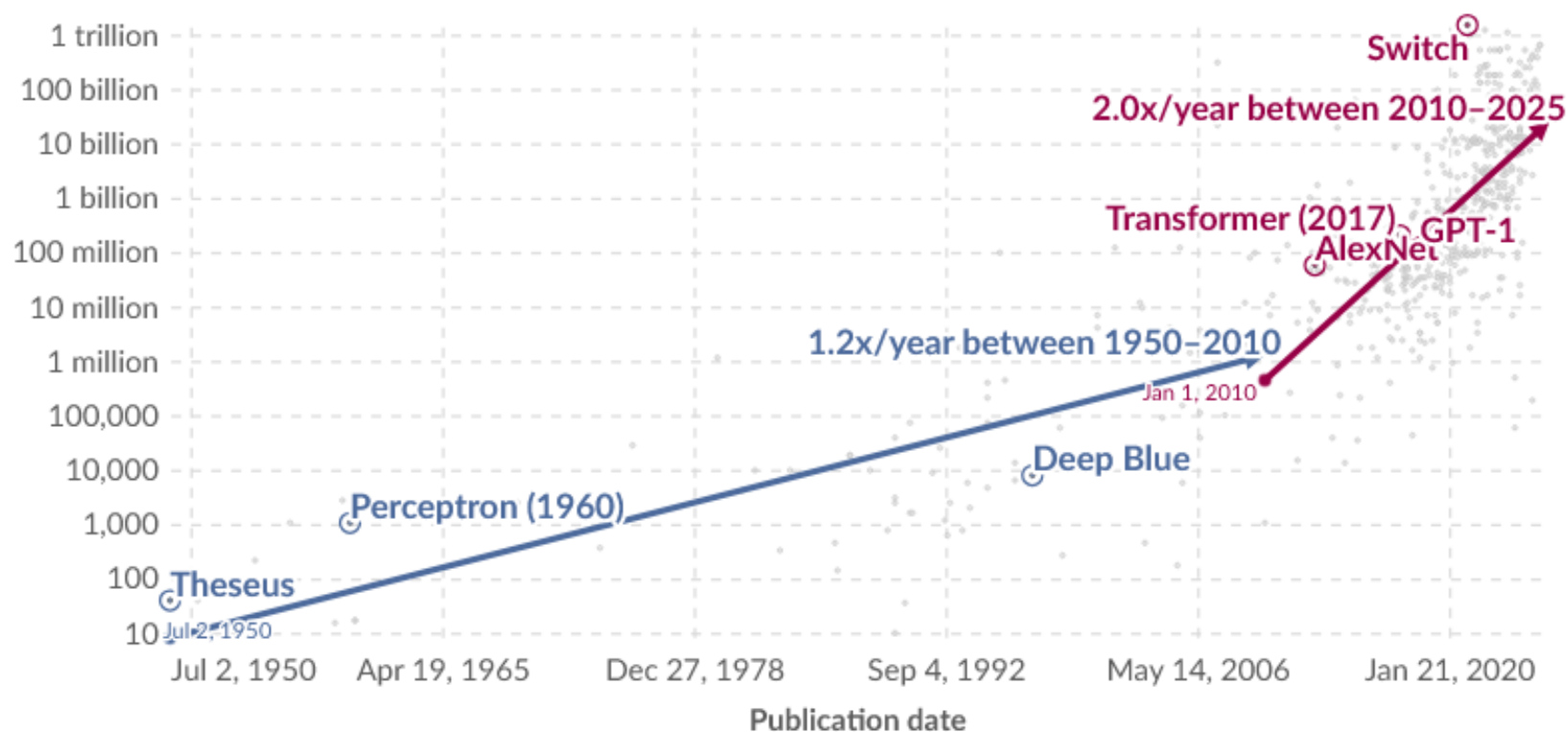
Parameters are variables in an AI system whose values are adjusted during training to establish how input data gets transformed into the desired output; for example, the connection weights in an artificial neural network.

Table Chart

Select systems

Settings

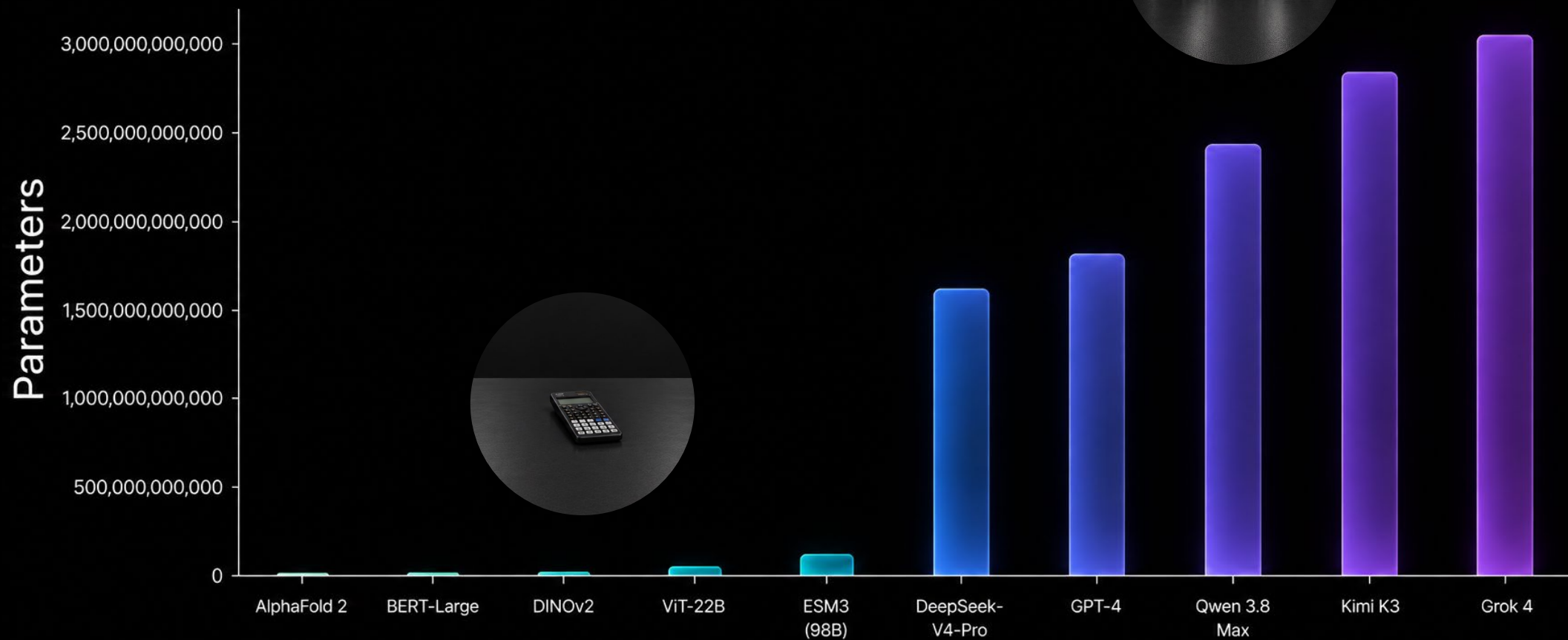
Number of parameters



Jul 2, 1950



Mar 6, 2025



Large Language Models (General-Purpose)

Obtain the data

- What type of data will be needed?
- Is the data sensitive?
- How much data will be needed?
- What problems do you foresee?
- Are there commonly used benchmarking datasets?

Model evaluation

Splitting data

- **Training set:** to train models
- **Validation set (Development set):** to evaluate trained models during development
- **Test set:** to assess generalization after development
- Additional independent test data is desirable



Evaluation metrics

- Primary performance metric
 - Must match research goal
 - Used to rank models
- Secondary performance metric(s)
 - May serve to exclude models
 - May be used to select between models similar in primary metric

Many different metrics and evaluation tools exist!

Model evaluation: Scatter plots – True vs Predicted



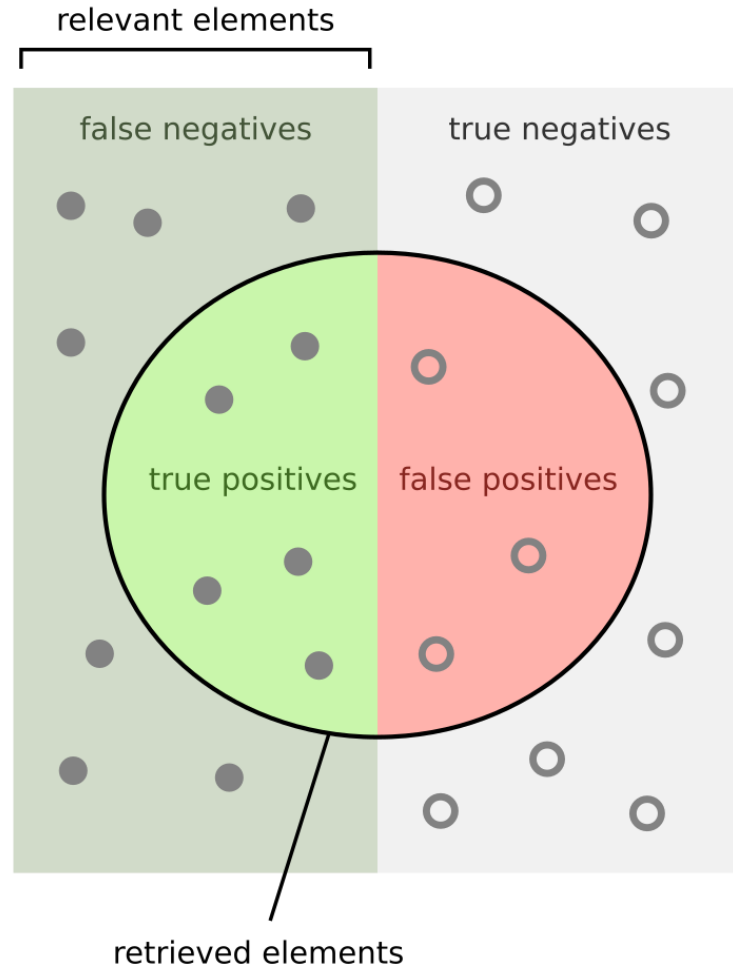
Calculate
correlation!

Model evaluation: Confusion matrix

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Total population = P + N		
	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

		Predicted condition	
		Cancer	Non-cancer
Actual condition	Total 8 + 4 = 12	7	5
	Cancer 8	6	2
	Non-cancer 4	1	3

Model evaluation: Precision and Recall

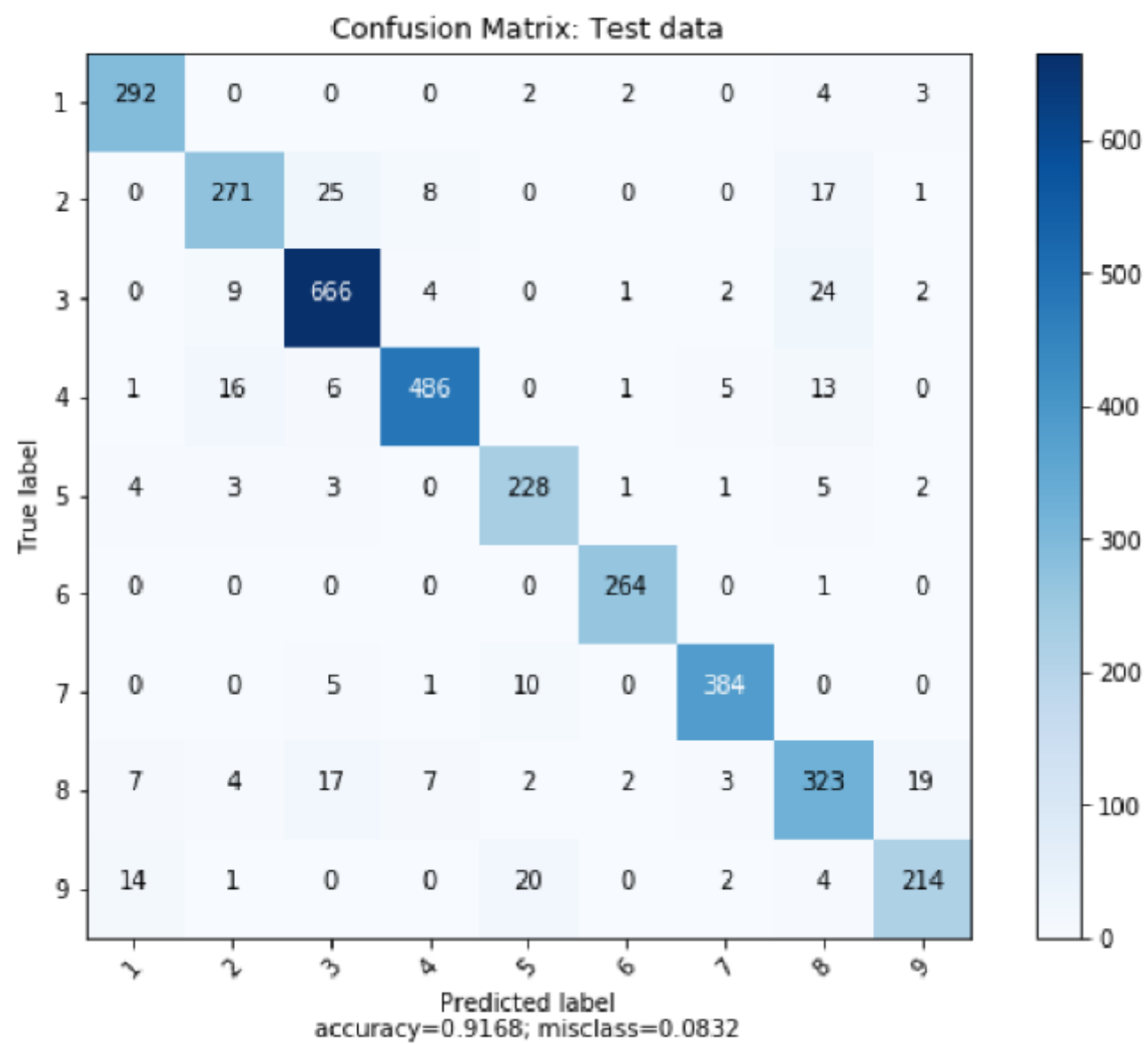


How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$



Model evaluation:
F1 score

$$F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Done training?

Final evaluation

- Evaluate on test set to assess generalization
- Use additional independent test set(s), if possible established benchmarking sets (compare to current SOTA)
- Compare to human performance (if relevant)
- Evaluate additional operational requirements

Expectation:

Test error > Validation error > Training error
(but ideally as little difference as possible)

Additional operational requirements

- Runtime
- Hardware requirements (e.g. memory, GPU)
- Explainability
- ...

Deployment

- Embedding AI model into a larger system which includes all steps from data input to final output and attractive user interface
- Incorporation into existing workflows/infrastructure
- Documentation
- External evaluation/approval
- Training of users

Continuous monitoring

- Changing data patterns will affect model performance
- Examples not represented in training data may not be handled as desired
- New demands/technology can arise
 - Continuous monitoring required
 - Model replacement/retraining when needed

We need to evaluate people, too.

Summary

- Choose the right AI for the problem
- Clear goals and high-quality data are essential for success
- Evaluation metrics need to fit research goal
- Avoid “data leakage” between split subsets: should be independent
- Test set needs to be held back to assess generalization
- Development continues after model training for deployment and monitoring

Beyond this course...

How to learn more about AI

- On-site courses
- Online courses
- Blogs
- Youtube
- Books
- Collaborate with pros
- AI as a teacher

<https://github.com/Aitslab/training>

Aitslab / training

Watch ▾

0

★ Star

0

🔗 Fork

0

<> Code

! Issues 0

🔗 Pull requests 0

📁 Projects 0

📖 Wiki

🛡 Security

📊 Insights

⚙ Settings

Branch: master ▾

training / README.md

Find file

Copy path

 SonjaAits Update README.md

830a97f now

1 contributor

234 lines (109 sloc) | 4.41 KB

Raw

Blame

History



Training resources

Training materials related to data science and artificial intelligence. Resources which are not available for free are marked (\$). You can find links to organizations which provide physical courses (in physicalcourses.md) and links to data sources (in datasources.md)

<https://github.com/COMPUTE-LU/>

Popular repositories

PLGroup_AI4MedLife-journalclub

Public

info for the COMPUTE peer-learning group "AI for Medicine and Life Sciences - Journal Club"

☆ 6

AI4MedLife_imaging_2021

Public

a repo for the course "AI for Medicine and Life Sciences - AI for Image and Video Data" (fall term 2021) at the COMPUTE research school, Lund University (<https://www.compute.lu.se/>).

● Jupyter Notebook ☆ 3 🍴 3

PLGroup_PythonforBeginners

Public

material for and from the peer-learning group Python for Beginners

● Jupyter Notebook

jupyter-course

Public

Forked from [mlund/jupyter-course](#)

Jupyter Course

● Jupyter Notebook 🍴 1

.github

Public

overview page for COMPUTE research school

AI4MedLife_intro_2023

Public

repo for the PhD level course Artificial Intelligence in Medicine and Life Sciences -Introduction (NTF012F) at Lund University, Dec 2023

Lund has a rich environment of organizations supporting AI in medicine and life sciences

COMPUTE PhD School: <https://www.compute.lu.se/>

AI Lund: <http://ai.lu.se/>

LU profile area Natural and Artificial Cognition

LU profile area Nature-based Future Solutions

LTH profile area AI and Digitalization

LTH profile area Engineering Health

WASP/WASP-HS/DDLS

www.compile.lu.se